

Question 41: In an organization, number of departments exists. Each department has a name & unique code. Number of employees work in each department. Each employee has unique employee code. Detailed information like name, address, city, basic, date of join are also stored. In a leave register for each employee leave records are kept showing leave type (CL/EL/ML etc.), from-date and to-date. When an employee retires or resigns then all the leave information pertaining to him are also deleted. Basic salary must be within Rs.5000 to Rs.9000. A department can not be deleted if any employee record refers to it. Valid grades are A/B/C. Employee name must be in uppercase only. Default value for joining date is system date. Design & implement the tables with necessary constraints to support the scenario depicted above.

Query :

```
CREATE TABLE Department(  
    dcode VARCHAR(10) PRIMARY KEY,  
    dname VARCHAR(50) NOT NULL UNIQUE,  
    no_of_employee INT DEFAULT 0  
);  
  
CREATE TABLE Employee(  
    empcode INT PRIMARY KEY,  
    ename VARCHAR(50) NOT NULL,  
    address VARCHAR(100),  
    city VARCHAR(50),  
    basic DECIMAL(8,2),  
    doj DATE DEFAULT (CURRENT_DATE),  
    grade CHAR(1),  
    dcode VARCHAR(10),  
  
    CONSTRAINT chk_basic CHECK (basic BETWEEN 5000 AND 9000),  
    CONSTRAINT chk_grade CHECK (grade IN ('A','B','C')),  
    CONSTRAINT chk_uppercase CHECK (ename = UPPER(ename)),  
    CONSTRAINT fk_emp_dept FOREIGN KEY (dcode) REFERENCES Department(dcode) ON  
DELETE RESTRICT  
);  
  
CREATE TABLE LeaveRegister(  
    leave_id INT AUTO_INCREMENT PRIMARY KEY,  
    empcode INT NOT NULL,  
    leave_type VARCHAR(10) NOT NULL,  
    from_date DATE NOT NULL,  
    to_date DATE NOT NULL,  
  
    CONSTRAINT fk_leave_emp FOREIGN KEY(empcode) REFERENCES Employee(empcode)  
ON DELETE CASCADE  
);
```

Output :

```
mysql> CREATE TABLE Department(  
->    dcode VARCHAR(10) PRIMARY KEY,  
->    dname VARCHAR(50) NOT NULL UNIQUE,  
->    no_of_employee INT DEFAULT 0  
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> CREATE TABLE Employee(  
->     empcode INT PRIMARY KEY,  
->     ename VARCHAR(50) NOT NULL,  
->     address VARCHAR(100),  
->     city VARCHAR(50),  
->     basic DECIMAL(8,2),  
->     doj DATE DEFAULT (CURRENT_DATE),  
->     grade CHAR(1),  
->     dcode VARCHAR(10),  
->  
->     CONSTRAINT chk_basic CHECK (basic BETWEEN 5000 AND 9000),  
->     CONSTRAINT chk_grade CHECK (grade IN ('A','B','C')),  
->     CONSTRAINT chk_uppercase CHECK (ename = UPPER(ename)),  
->     CONSTRAINT fk_emp_dept FOREIGN KEY (dcode) REFERENCES Department(dcode) ON DELETE RESTRICT  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE LeaveRegister(  
->     leave_id INT AUTO_INCREMENT PRIMARY KEY,  
->     empcode INT NOT NULL,  
->     leave_type VARCHAR(10) NOT NULL,  
->     from_date DATE NOT NULL,  
->     to_date DATE NOT NULL,  
->  
->     CONSTRAINT fk_leave_emp FOREIGN KEY(empcode) REFERENCES Employee(empcode) ON DELETE CASCADE  
-> );  
Query OK, 0 rows affected (0.02 sec)
```

Question 42: Try to violate the constraints that you have implemented in the table & note, what happens. [Try with suitable INSERT/UPDATE/DELETE instruction]

Query :

```
INSERT INTO Employee(empcode,ename,address,city,basic,grade,dcode) VALUES
(104,'RAKESH','ABC','KOLKATA',10000,'A','D1');
```

```
INSERT INTO Employee (empcode,ename,address,city,basic,grade,dcode) VALUES
(105,'RAMESH','ABC','KOLKATA',7000,'D','D1');
```

```
INSERT INTO Employee(empcode,ename,address,city,basic,grade,dcode) VALUES
(107,'SOUMEN','ABC','KOLKATA',7000,'A','D10');
```

```
DELETE FROM Department
WHERE dcode='D1';
```

```
DELETE FROM Employee
WHERE empcode=101;
```

Output :

```
mysql> INSERT INTO Employee(empcode,ename,address,city,basic,grade,dcode) VALUES
-> (104,'RAKESH','ABC','KOLKATA',10000,'A','D1');
ERROR 3819 (HY000): Check constraint 'chk_basic' is violated.
```

```
mysql> INSERT INTO Employee (empcode,ename,address,city,basic,grade,dcode) VALUES
-> (105,'RAMESH','ABC','KOLKATA',7000,'D','D1');
ERROR 3819 (HY000): Check constraint 'chk_grade' is violated.
```

```
mysql> INSERT INTO Employee(empcode,ename,address,city,basic,grade,dcode) VALUES
-> (107,'SOUMEN','ABC','KOLKATA',7000,'A','D10');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
(`assignments`.`employee`, CONSTRAINT `fk_emp_dept` FOREIGN KEY (`dcode`) REFERENCES
`department` (`dcode`) ON DELETE RESTRICT)
```

```
mysql> DELETE FROM Department
-> WHERE dcode='D1';
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key
constraint fails (`assignments`.`employee`, CONSTRAINT `fk_emp_dept` FOR
EIGN KEY (`dcode`) REFERENCES `department` (`dcode`) ON DELETE RESTRICT)
mysql>
```

```
mysql> DELETE FROM Employee
-> WHERE empcode=101;
Query OK, 1 row affected (0.00 sec)
```

Question 43:

a) create a view showing employee code, name, dcode & Basic For a particular department.

Query :

```
CREATE VIEW D1_EMP_VIEW AS
```

```
SELECT empcode, ename, dcode, basic FROM Employee WHERE dcode='D1';
```

Output :

```
mysql> CREATE VIEW D1_EMP_VIEW AS
-> SELECT empcode, ename, dcode, basic FROM Employee WHERE dcode='D1';
Query OK, 0 rows affected (0.00 sec)
```

b) Try to ensure a row into the view with valid department & also with invalid ones.

Query :

```
INSERT INTO D1_EMP_VIEW VALUES(201, 'ANIK', 'D1', 7500);
```

```
INSERT INTO D1_EMP_VIEW VALUES(202, 'ROHIT', 'D5', 7000);
```

Output :

```
mysql> INSERT INTO D1_EMP_VIEW VALUES
mysql> INSERT INTO D1_EMP_VIEW VALUES
-> (202, 'ROHIT', 'D5', 7000);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
(`assignments`.`employee`, CONSTRAINT `fk_emp_dept` FOREIGN KEY (`dcode`) REFERENCES
`department` (`dcode`) ON DELETE RESTRICT)
```

c) Find the newly inserted row in the table From which view was created .

Query :

```
SELECT * FROM Employee WHERE empcode=201;
```

Output :

```
mysql> SELECT * FROM Employee
-> WHERE empcode=201;
+-----+-----+-----+-----+-----+-----+-----+-----+
| empcode | ename | address | city | basic | doj | grade | dcode |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 201 | ANIK | NULL | NULL | 7500.00 | NULL | NULL | D1 |
+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

d) Try to increment basic by Rs.100/-

Query :

```
UPDATE D1_EMP_VIEW SET basic = basic + 100;
```

Output :

```
mysql> UPDATE D1_EMP_VIEW
-> SET basic = basic + 100;
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2 Changed: 2 Warnings: 0
```

e) Check it in the original table.

Query :

```
SELECT empcode, ename, basic FROM Employee;
```

Output :

```
mysql> SELECT empcode, ename, basic
-> FROM Employee;
+-----+-----+-----+
| empcode | ename | basic |
+-----+-----+-----+
| 102 | AMIT | 6600.00 |
| 103 | SUMAN | 8000.00 |
| 201 | ANIK | 7600.00 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

f) Delete the view.

Query :

```
DROP VIEW D1_EMP_VIEW;
```

Output :

```
mysql> DROP VIEW D1_EMP_VIEW;
Query OK, 0 rows affected (0.01 sec)
```


Question 44:

a) create a view Showing empcode, name, deptname, basic, leave type, From date & to date.

Query :

```
CREATE VIEW EMP_LEAVE_VIEW AS
SELECT E.empcode, E.ename, D.dname, E.basic, L.leave_type, L.from_date,
L.to_date
FROM Employee E JOIN Department D ON E.dcode = D.dcode
JOIN LeaveRegister L ON E.empcode = L.empcode;
```

Output :

```
mysql> CREATE VIEW EMP_LEAVE_VIEW AS
-> SELECT E.empcode, E.ename, D.dname, E.basic, L.leave_type, L.from_date, L.to_date
-> FROM Employee E JOIN Department D ON E.dcode = D.dcode
-> JOIN LeaveRegister L ON E.empcode = L.empcode;
Query OK, 0 rows affected (0.03 sec)
```

b) Try to insert a row in the view. Check what happens?

Query :

```
INSERT INTO EMP_LEAVE_VIEW VALUES
(500, 'JOY', 'COMPUTER SCIENCE', 7000, 'CL', '2026-04-01', '2026-04-03');
```

Output :

```
mysql> INSERT INTO EMP_LEAVE_VIEW VALUES
-> (500, 'JOY', 'COMPUTER SCIENCE', 7000, 'CL', '2026-04-01', '2026-04-03');
ERROR 1394 (HY000): Can not insert into join view 'assignments.emp_leave_view' without fields list
mysql>
```

c) Try to increment basic by Rs.100.

Query :

```
UPDATE EMP_LEAVE_VIEW
SET basic = basic + 100;
```

Output :

```
mysql> UPDATE EMP_LEAVE_VIEW
-> SET basic = basic + 100;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

d) Delete the view.

Query :

```
DROP VIEW EMP_LEAVE_VIEW;
```

Output :

```
mysql> DROP VIEW EMP_LEAVE_VIEW;
Query OK, 0 rows affected (0.00 sec)
```

Question 45:

a) Create a table having empcode , Name, deptname, & basic From the existing tables along with the records of the employee who are in a particular department (say, d1) and with a basic Rs. 7000/-

Query :

```
CREATE TABLE D1_EMPLOYEE AS
SELECT E.empcode, E.ename, D.dname, E.basic
FROM Employee E JOIN Department D ON E.dcode = D.dcode
WHERE E.dcode='D1' AND E.basic=7000;
```

Output :

```
mysql> CREATE TABLE D1_EMPLOYEE AS
-> SELECT E.empcode, E.ename, D.dname, E.basic
-> FROM Employee E JOIN Department D ON E.dcode = D.dcode
-> WHERE E.dcode='D1' AND E.basic=7000;
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

b) From the existing table, add the employees with the basic salary greater than or equal to 7000/-

Query :

```
INSERT INTO D1_EMPLOYEE
SELECT E.empcode, E.ename, D.dname, E.basic
FROM Employee E JOIN Department D ON E.dcode = D.dcode
WHERE E.basic >= 7000;
```

Output :

```
mysql> INSERT INTO D1_EMPLOYEE
-> SELECT E.empcode, E.ename, D.dname, E.basic
-> FROM Employee E JOIN Department D ON E.dcode = D.dcode
-> WHERE E.basic >= 7000;
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

c) Alter the table to add a net pay column.

Query :

```
ALTER TABLE D1_EMPLOYEE
ADD netpay DECIMAL(10,2);
```

Output :

```
mysql> ALTER TABLE D1_EMPLOYEE
-> ADD netpay DECIMAL(10,2);
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

d) Replace net pay with 1.5* Basic.

Query :

```
UPDATE D1_EMPLOYEE
SET netpay = basic * 1.5;
```

Output :

```
mysql> UPDATE D1_EMPLOYEE
-> SET netpay = basic * 1.5;
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2 Changed: 2 Warnings: 0
```

empcode	ename	dname	basic	netpay
103	SUMAN	ELECTRONICS	8000.00	12000.00
201	ANIK	COMPUTER SCIENCE	7600.00	11400.00

```
mysql> SELECT * FROM D1_EMPLOYEE;
2 rows in set (0.00 sec)
```

e) Try to remove the net net pay column. [It may require no. of steps]

Query :

```
ALTER TABLE D1_EMPLOYEE
DROP COLUMN netpay;
```

Output :

```
mysql> ALTER TABLE D1_EMPLOYEE
-> DROP COLUMN netpay;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Question 46: Drop all the tables that you have created .

Query :

```
DROP TABLE LeaveRegister;
```

```
DROP TABLE Employee;
```

```
DROP TABLE Department;
```

```
DROP TABLE D1_EMPLOYEE;
```

Output :

```
mysql> DROP TABLE LeaveRegister;  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DROP TABLE Employee;  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DROP TABLE Department;  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DROP TABLE D1_EMPLOYEE;  
Query OK, 0 rows affected (0.01 sec)
```